

# AMIRALI AALI

M.Sc. Data Science Student | RWTH Aachen University

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## PROFILE

M.Sc. Data Science student at RWTH Aachen with 4+ years of industry experience across computer vision, machine learning, embedded systems, cloud data engineering, and privacy engineering. Experience developing real-time ML systems for automotive applications at Aptiv and Porsche and large-scale data pipelines at SAP.

## TECHNICAL EXPERIENCE

### Working Student – Vision-Based Driving Assistance Systems

Porsche

Aug 2026 – Mar 2027 | 8 mos

Weissach, Germany

- Currently working on data-centric development of ML models.

### Intern – Vision-Based Driving Assistance Systems

Porsche

Apr 2026 – Jul 2026 | 3 mos

Weissach, Germany

- Worked on data-centric development of ML models.

### Working Student – Data De-Identification

SAP

Apr 2024 – Mar 2026 | 2 yrs

Düsseldorf-Ratingen, Germany

- Part of the anonymization team delivering enterprise-scale data de-identification services.
- Designed and maintained Databricks pipelines integrated with Azure Data Lake Storage (ADLS), implementing Python-based text anonymization workflows compliant with data protection requirements.
- Automated pipeline processing enterprise-scale datasets with PySpark/Databricks on Azure ADLS

### Bachelor's Thesis Student

Aptiv

Jan 2024 – Mar 2024 | 3 mos

Wuppertal, Germany

- Thesis: *Street Lamp Area Detection* – a joint project with the University of Duisburg-Essen combining classical vision and CNN inference.

### Working Student – Computer Vision / AI / ML

Aptiv

Nov 2022 – Dec 2023 | 1 yr 2 mos

Wuppertal, Germany

- Collaborated with a specialist team on a deep-learning high-beam assistant for Gen. 7 automotive systems.
- Built a real-time smart controller that automatically adapts high-beam intensity based on scene detection.
- Designed real-time CNN inference module for vehicle embedded controllers

## EDUCATION

### M.Sc. Data Science

RWTH Aachen University

Apr 2024 – Present

Aachen, Germany

Thesis: *RL-Guided Exploration for Learning Lifted Action Schemas with DIAS*

### B.Sc. Software Engineering

University of Duisburg-Essen

Oct 2020 – Mar 2024

Duisburg, Germany

Thesis: *Street Lamp Area Detection using Classical Vision and CNNs* | GPA: **1.7** (very good, German scale)

## PROJECTS

### Rapid Time Series Datasets Library with Rust – Rust, Python, NumPy, Gitlab

Python package backed by Rust for efficient preprocessing of time series data, integrating with PyTorch via the LightningModule interface. [github.com/amiraliaali/rapid\\_time\\_series\\_dataset\\_library](https://github.com/amiraliaali/rapid_time_series_dataset_library)

### Face Anonymizer in Rust – Rust, Python, PyBinds, Computer Vision

High-performance face anonymization pipeline with Python bindings, enabling real-time processing.

## SKILLS

**Data Science & ML:** Python, PyTorch, Computer Vision, OpenCV, NumPy, Pandas, Scikit-Learn, Reinforcement Learning

**Systems & Low-Level:** C/C++, Rust, PyBinds, Bazel

**Data & Engineering:** Git/GitHub, Gerrit, Linux, Azure, DataBricks, PySpark, ADLS

## LANGUAGES & HONOURS

**Languages:** English (C1) | German (B2) | Persian (Native)

**German National Scholarship** – Deutschlandstipendium, RWTH Aachen (Sponsor: Dr. Ing. h.c. F. Porsche AG), 2024–2025